Practice Handout: Proving Θ Runtime Bounds (for Beginners)

Overview

This handout introduces how to prove that an algorithm's runtime (or a function) is in $\Theta(g(n))$. The goal is to understand what it means for a function to have both an upper and lower asymptotic bound.

Definition Reminder

For functions $f(n)$ and $g(n)$, we say:

$$f(n) \in \Theta(g(n)) \iff \exists c_1, c_2 > 0,\, n_0 : \ \forall n \geq n_0,\ c_1 g(n) \leq f(n) \leq c_2 g(n).$$

In words:

- $f(n)$ grows at the *same rate* as $g(n)$, up to constant factors.
- The constants c_1 and c_2 are "safety bounds."
- n_0 is the point beyond which both inequalities hold.

Visual intuition: $f(n)$ is "sandwiched" between $c_1 g(n)$ and $c_2 g(n)$ for large enough n.

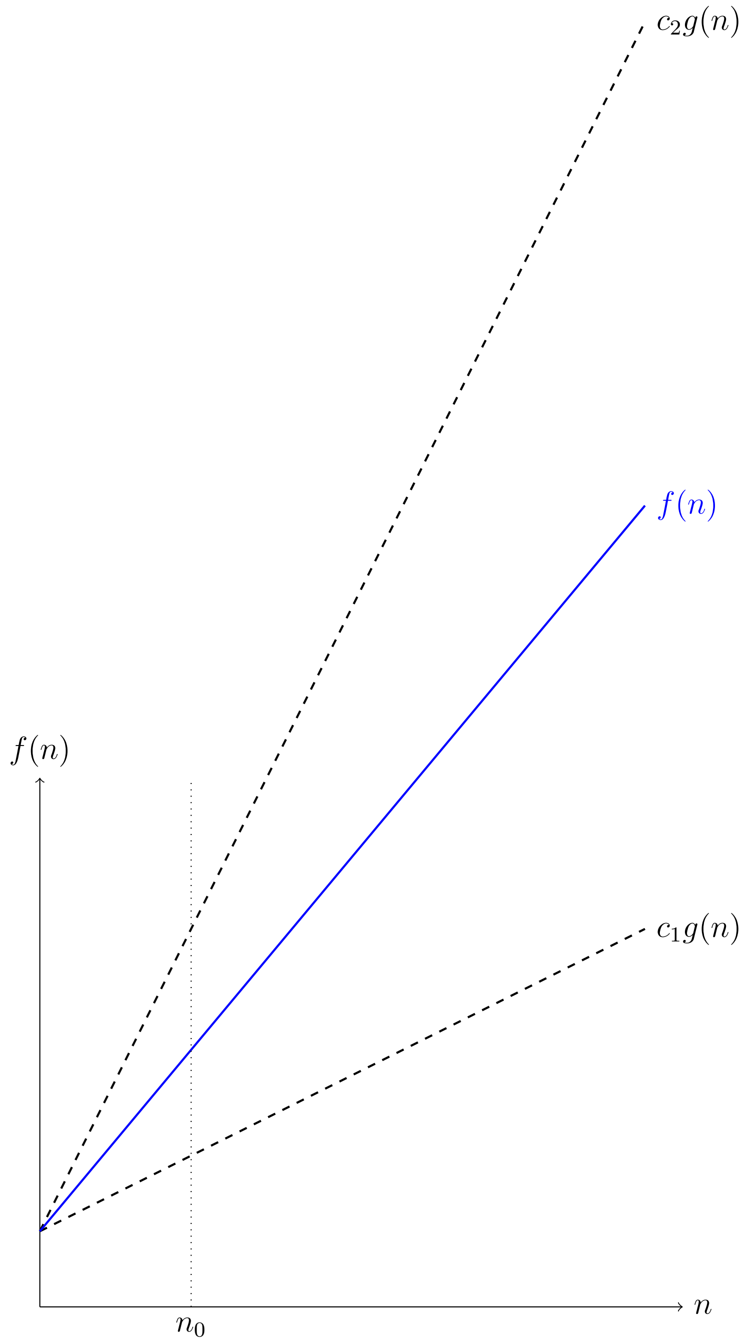

Figure 1: Intuition for Θ notation.

How to Prove a Θ Bound (Steps)

1. Identify the **dominant term** of $f(n)$ for large n.
2. Guess $g(n)$ as that dominant term.

3. Show constants c_1 and c_2 such that $c_1 g(n) \le f(n) \le c_2 g(n)$ for $n \ge n_0$.
4. (Optional) Verify graphically or by simple algebra.

Example Problem

Show that $f(n) = 3n + 5 \in \Theta(n)$**.**
For $n \ge 5$:

$$3n \le 3n + 5 \le 4n.$$

Thus, with $c_1 = 3$, $c_2 = 4$, and $n_0 = 5$, we have $f(n) \in \Theta(n)$.

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Practice Problems

Each of the following problems asks you to prove that $f(n)$ belongs to $\Theta(g(n))$. Show your reasoning and identify constants c_1, c_2, and n_0.

Set A: Linear and Polynomial Growth

A1. Prove $f(n) = 7n + 9 \in \Theta(n)$.

A2. Prove $f(n) = 5n^2 + 10n + 20 \in \Theta(n^2)$.

A3. Prove $f(n) = 4n^3 + 3n^2 + 2n + 1 \in \Theta(n^3)$.

A4. Prove $f(n) = 2n^4 - n^3 + 10n \in \Theta(n^4)$.

Set B: Logarithmic and Superlinear Growth

B1. Show $f(n) = 4 \log_2 n + 10 \in \Theta(\log n)$.

B2. Show $f(n) = 3n \log n + 20n \in \Theta(n \log n)$.

B3. Show $f(n) = 2n^2 + 50n \log n \in \Theta(n^2)$.

Set C: Exponential and Mixed Growth

C1. Show $f(n) = 3^n + 2n^5 \in \Theta(3^n)$.

C2. Show $f(n) = n^3 + 2^{n/2} \in \Theta(2^{n/2})$.

C3. Show $f(n) = 10^n + 5^n \in \Theta(10^n)$.

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Quick Hints for Students

- When multiple terms are added, the one that grows fastest determines Θ.
- Ignore constant multipliers and lower-order terms.
- If logarithms are involved, check the base but remember that $\log_a n \in \Theta(\log_b n)$ for any $a, b > 1$.
- For exponentials, a^n dominates b^n when $a > b$.
- Always show the bounding inequalities at least once to build intuition.

Challenge Extension

For extra practice, try proving:

$$f(n) = n^3 + n^2 \log n + 100n \in \Theta(n^3)$$

and sketch a simple justification using inequalities or limits.

Further Reading

Cormen, Leiserson, Rivest, and Stein, *Introduction to Algorithms*, Chapter on Asymptotic Notation.

Dasgupta, Papadimitriou, and Vazirani, *Algorithms*, Appendix on Growth of Functions.